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IMO 2009 Day-1

1. Let n be a positive integer and let $a_1, a_2, \dots, a_k$ ($k \geq 2$) be distinct integers in $\{1, \dots, n\}$ such that n divides $a_i(a_{i+1} - 1)$ for $i = 1, \dots, k - 1$. Prove that n does not divide $a_k(a_1 - 1)$.
2. Let ABC be a triangle with circumcentre O . Points P and Q are interior points of sides CA and AB respectively. Let K, L, M be the midpoints of BP, CQ, PQ respectively. Let Γ be the circle passing through K, L, M . Suppose PQ is tangent to Γ . Prove that $OP = OQ$.
3. Suppose that $s_1, s_2, s_3, \dots$ is a strictly increasing sequence of positive integers such that the subsequences $s_1, s_3, s_5, \dots$ and $s_2, s_4, s_6, \dots$ are arithmetic progressions. Prove that $s_1, s_2, s_3, \dots$ is an arithmetic progression.

Day 2

4. Let ABC be a triangle with $AB = AC$. The angle bisectors of $\angle CAB$ and $\angle ABC$ meet sides BC and CA at D and E respectively. Let K be the incenter of triangle ADC . Suppose $\angle BEK = 45^\circ$. Find all possible values of $\angle CAB$.
5. Determine all functions f from the set of positive integers to itself such that for all positive integers a and b , there exists a non-degenerate triangle with sides a , $f(b)$ and $f(a) - 1$.
6. Let $a_1, a_2, \dots, a_n$ be distinct positive integers and let M be a set of $n - 1$ positive integers containing $s = a_1 + a_2 + \dots + a_n$. A grasshopper starts at 0 on the real axis and makes jumps of lengths $a_1, a_2, \dots, a_n$ in some order. Prove that the order can be chosen so that the grasshopper never lands on any point of M .